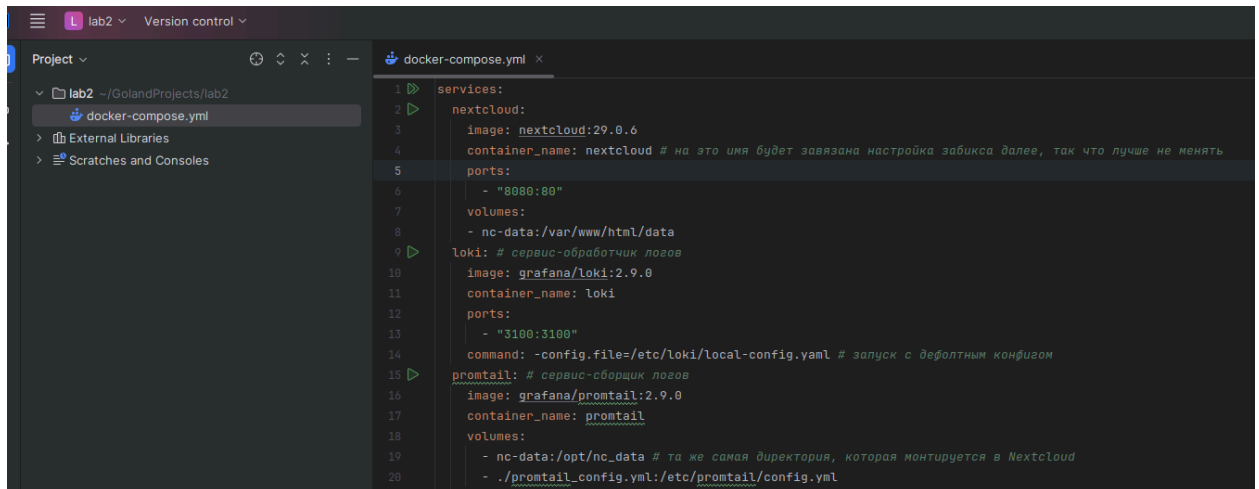
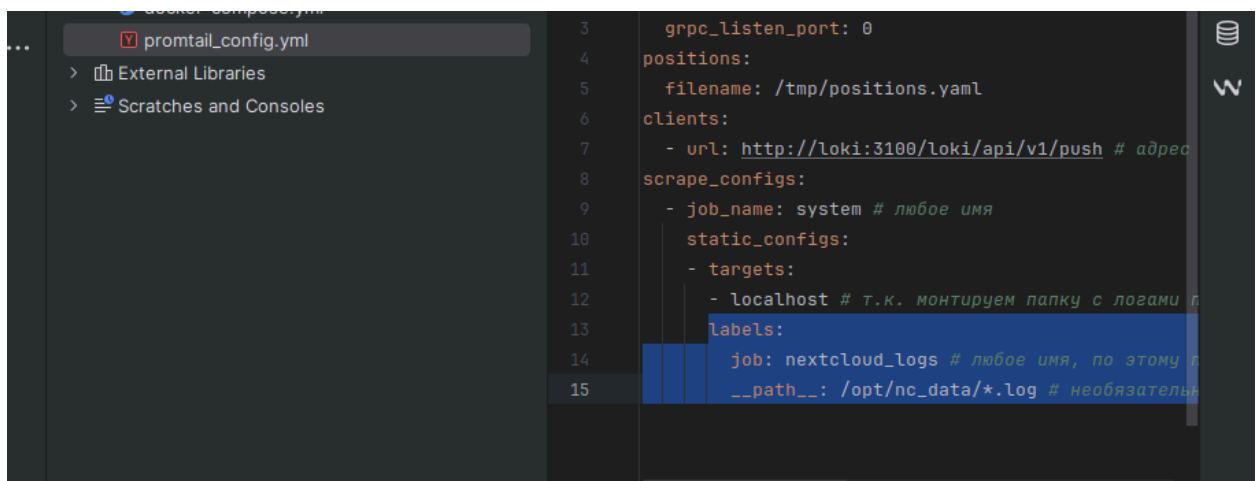


1. Создадим и заполним docker-compose.yml. Пропишем настройки запуска для тестового инстанса NextCloud, PromTail (сборщик логов), Loki (обработчик логов), Grafana (Визуализация логов/метрик) и самого Zabbix (система мониторинга – ему нужен фронт, бэк и база).



```
1 services:
2   nextcloud:
3     image: nextcloud:29.0.6
4     container_name: nextcloud # на это имя будет завязана настройка зabbix далее, так что лучше не менять
5     ports:
6       - "8080:80"
7     volumes:
8       - nc-data:/var/www/html/data
9   loki: # сервис-обработчик логов
10    image: grafana/loki:2.9.0
11    container_name: loki
12    ports:
13      - "3100:3100"
14    command: -config.file=/etc/loki/local-config.yaml # запуск с дефолтным конфигом
15   promtail: # сервис-сборщик логов
16    image: grafana/promtail:2.9.0
17    container_name: promtail
18    volumes:
19      - nc-data:/opt/nc_data # та же самая директория, которая монтируется в Nextcloud
20      - ./promtail_config.yaml:/etc/promtail/config.yaml
```

2. Заполним файл конфигурации PromTail. Укажем порт http, адрес Loki и где сами log файлы следует искать.



```
3   grpc_listen_port: 0
4   positions:
5     filename: /tmp/positions.yaml
6   clients:
7     - url: http://loki:3100/loki/api/v1/push # адрес
8   scrape_configs:
9     - job_name: system # любое имя
10      static_configs:
11        - targets:
12          - localhost # т.к. монтируем папку с логами
13          labels:
14            job: nextcloud_logs # любое имя, по этому
15            __path__: /opt/nc_data/*.log # необязательно
```

3. Сделаем Docker Compose Up

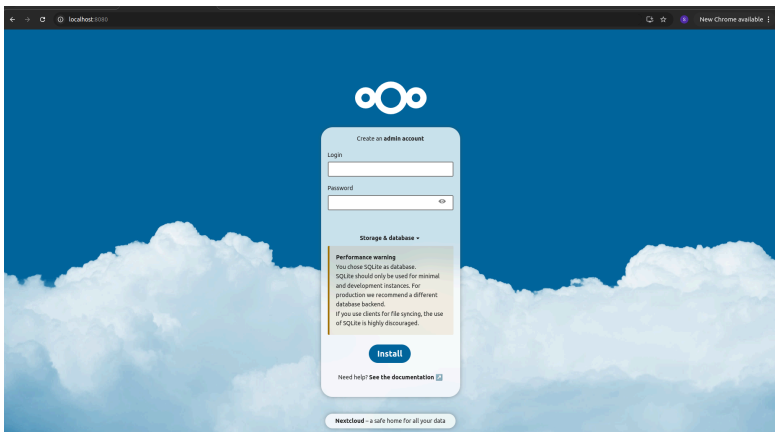
```
and is the expected type
undercat@undercat-Aspire-A315-59:~/GoLandProjects/Lab2$ docker compose up
[+] Running 4/4
  ✓ Container loki          Running      0.0s
  ✓ Container postgres-zabbix Running    0.0s
  ✓ Container grafana       Running    0.0s
  ✓ Container nextcloud     Running    0.0s
Attaching to grafana, loki, nextcloud, postgres-zabbix, promtail, zabbix-back, zabbix-front
zabbix-front | ** Deploying Zabbix web-interface (Nginx) with PostgreSQL database
zabbix-front | ** Using POSTGRES_USER variable from ENV
zabbix-front | ** Using POSTGRES_PASSWORD variable from ENV
zabbix-front | *****
zabbix-front | * DB_SERVER_HOST: postgres-zabbix
zabbix-front | * DB_SERVER_PORT: 5432
zabbix-front | * DB_SERVER_PORT: 5432
zabbix-front | * DB_SERVER_DBNAME: zabbix
zabbix-front | * DB_SERVER_SCHEMA: public
zabbix-front | *****
zabbix-back  | ** Preparing Zabbix server
zabbix-back  | ** Preparing database
zabbix-back  | ** Using POSTGRES_USER variable from ENV
```

Какие-то логи сыпятся. На всякий проверим статусы контейнеров.

CONTAINER ID	IMAGE	COMMAND	NAMES	CREATED	STATUS	PORTS
44325c3c9451	zabbix/zabbix-web-nginx-pgsql:ubuntu-6.4-latest	"docker-entrypoint.sh"	zabbix-front	2 minutes ago	Up 2 minutes (healthy)	8443/tcp, 0.0.0.0:8082->8080/tcp, [::]:8082->8080/tcp
c27b0a8488df	zabbix/zabbix-server-pgsql:ubuntu-6.4-latest	"/usr/bin/tini -- /u..."	zabbix-back	2 minutes ago	Up 2 minutes	0.0.0.0:10051->10051/tcp, [::]:10051->10051/tcp
90e738831257	nextcloud:29.0.6	"/entrypoint.sh apac..."	nextcloud	2 minutes ago	Up 2 minutes	0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
5a4a44a77b98	grafana/grafana:11.2.0	"/run.sh /run.sh"	grafana	2 minutes ago	Up 2 minutes	0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp
52a43ceb0b81	postgres:15	"docker-entrypoint.s..."	postgres-zabbix	2 minutes ago	Up 2 minutes (healthy)	5432/tcp
1dad6f803e3f	grafana/promtail:2.9.0	"/usr/bin/promtail -..."	promtail	2 minutes ago	Up 2 minutes	
da23fee7970b	grafana/loki:2.9.0	"/usr/bin/loki -conf..."	loki	2 minutes ago	Up 2 minutes	0.0.0.0:3100->3100/tcp, [::]:3100->3100/tcp

Можно быть уверенными, что инфру подняли.

4. На локальном порту 8080 откроем веб интерфейс NextCloud.



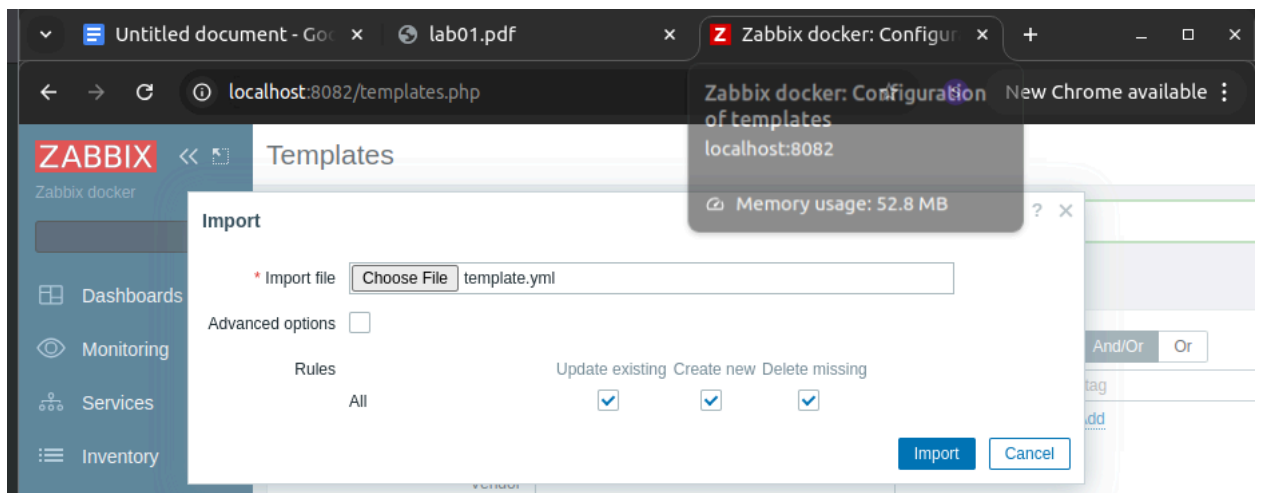
Проверим, что логи взаимодействия с NextCloud сыпятся куда нужно.

```
14-06/2/cp, 146/2/cp, 146/1-146/2/cp, 146/2/cp, 0-0-0-0-146/2-146/2/cp, 1-1-146/2-146/2/cp - upstream documents gateway reloading
undercat@undercat:~$ docker exec -it nextcloud /bin/sh
# ls
3rdparty  COPYING  composer.json  config  core  custom_apps  dist  index.php  nextcloud-init-sync.lock  ocs  package-lock.json  public.php  resources  status.php  version.php
AUTHORS  apps  composer.lock  console.php  cron.php  data  index.html  lib  occ  ocs-provider  package.json  remote.php  robots.txt  themes
# cd /var/www/html
# cat data/
cat: data/: Is a directory
# cd data
# ls
index.html  nextcloud.log
# cat nextcloud.log
{"reqId":"6sqQOWNP3qoIOXAeK09","level":2,"time":"2025-12-07T19:12:11+00:00","remoteAddr":"172.21.0.1","user":"","app":"no app in context","method":"GET","url":"/","message":"Host localhost was not connected to because it violates local access rules","userAgent":"Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/135.0.0.0 Safari/537.36","version":"","data":{}}
{"reqId":"6sqQOWNP3qoIOXAeK09","level":2,"time":"2025-12-07T19:12:11+00:00","remoteAddr":"172.21.0.1","user":"","app":"no app in context","method":"GET","url":"/","message":"Host localhost was not connected to because it violates local access rules","userAgent":"Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/135.0.0.0 Safari/537.36","version":"","data":{}}
{"reqId":"6sqQOWNP3qoIOXAeK09","level":5,"time":"2025-12-07T19:12:11+00:00","remoteAddr":"172.21.0.1","user":"","app":"PHP","method":"GET","url":"/","message":"fopen(/var/www/html/config/config.php): Failed to open stream: No such file or directory at /var/www/html/lib/private/config.php#221","userAgent":"Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/135.0.0.0 Safari/537.36","version":"","data":{"app":"PHP"}}
#
```

Проверим, что промтеил подбирает эти логи.

```
undercat@undercat:~$ docker logs promtail
level=info ts=2025-12-07T19:25:16.218195778Z caller=promtail.go:133 msg="Reloading configuration file" md5sum=2c8ba9ad5d476d9e9f64ddad5fed7eae
level=info ts=2025-12-07T19:25:16.221422048Z caller=server.go:322 http[::]:9090 grpc[::]:45349 msg="server listening on addresses"
level=info ts=2025-12-07T19:25:16.22214067Z caller=main.go:174 msg="Starting Promtail" version="(version=2.9.0, branch=HEAD, revision=2feb64f69)"
level=warn ts=2025-12-07T19:25:16.22243898Z caller=promtail.go:263 msg="enable watchConfig"
level=info ts=2025-12-07T19:25:21.220218942Z caller=filetargetmanager.go:361 msg="Adding target" key="/opt/nc_data/*_log.{job=\"nextcloud.logs\"}"
level=info ts=2025-12-07T19:25:21.228472302Z caller=filetarget.go:313 msg="watching new directory" directory=/opt/nc_data
ts=2025-12-07T19:25:21.228632755Z caller=log.go:168 level=info msg="Seeked /opt/nc_data/nextcloud.log - &{Offset:0 Whence:0}"
level=info ts=2025-12-07T19:25:21.228618273Z caller=tailer.go:145 component=tailer msg="tail routine: started" path=/opt/nc_data/nextcloud.log
level=info ts=2025-12-07T19:25:59.17428021Z caller=filetargetmanager.go:181 msg="received file watcher event" name=/opt/nc_data/htaccess.txt op=CREATE
level=info ts=2025-12-07T19:26:11.053606715Z caller=filetargetmanager.go:181 msg="received file watcher event" name=/opt/nc_data/htaccess.txt op=CREATE
level=info ts=2025-12-07T19:26:11.066436903Z caller=filetargetmanager.go:181 msg="received file watcher event" name=/opt/nc_data/owncloud.db op=CREATE
level=info ts=2025-12-07T19:26:11.070663963Z caller=filetargetmanager.go:181 msg="received file watcher event" name=/opt/nc_data/owncloud.db-journal op=CREATE
level=info ts=2025-12-07T19:26:11.082316258Z caller=filetargetmanager.go:181 msg="received file watcher event" name=/opt/nc_data/owncloud.db-wal op=CREATE
level=info ts=2025-12-07T19:26:11.082367921Z caller=filetargetmanager.go:181 msg="received file watcher event" name=/opt/nc_data/owncloud.db-shm op=CREATE
level=info ts=2025-12-07T19:26:15.194768248Z caller=filetargetmanager.go:181 msg="received file watcher event" name=/opt/nc_data/.ocdata op=CREATE
level=info ts=2025-12-07T19:26:15.194768248Z caller=filetargetmanager.go:181 msg="received file watcher event" name=/opt/nc_data/.ocdata op=CREATE
```

5. Подключимся к Zabbix, импортируем шаблон мониторинга.



6. Разрешим имя nextcloud в zabbix

```
undercat@undercat-Aspire-A315-59:~/GoLandProjects/lab2$ docker exec -u www-data -it nextcloud bash
www-data@90e738831257:~/html$ php occ config:system:set trusted_domains 1 --value="nextcloud"
System config value trusted_domains => 1 set to string nextcloud
www-data@90e738831257:~/html$
```

7. Добавим новый хост

☐	Name ▲	Items	Triggers	Graphs	Discovery	Web	Interface	Proxy	Templates	Status	Availability	Agent encryption	Info	Tags
☐	nextcloud	Items 1	Triggers 1	Graphs	Discovery	Web			Test ping template	Enabled		None		
☐	Zabbix server	Items 104	Triggers 59	Graphs 19	Discovery 5	Web	127.0.0.1:10050		Linux by Zabbix agent, Zabbix server health	Enabled	20x	None		

8. Перейдем в раздел мониторинга и увидим наш хелзчек

DATA						
With data Without data						
☐	Host ▲	Name	Last check	Last value	Change	Tags
☐	nextcloud	Nextcloud: ping service	24s	healthy		History
☐	Zabbix server	Zabbix server: Number of processed numeric (float) values per second	2s	0.6997	-0.01089	component: system Graph

9. Установим плагин zabbix для Графаны и рестартнем её

```
cc@undercat@undercat-Aspire-A315-59:~/GoLandProjects/lab2$ docker exec -it grafana bash -c "grafana cli plugins install alexanderzobnin-zabbix-app"
✓ Downloaded and extracted alexanderzobnin-zabbix-app v6.0.3 zip successfully to /var/lib/grafana/plugins/alexanderzobnin-zabbix-app

Please restart Grafana after installing or removing plugins. Refer to Grafana documentation for instructions if necessary.

undercat@undercat-Aspire-A315-59:~/GoLandProjects/lab2$ docker restart grafana
grafana
undercat@undercat-Aspire-A315-59:~/GoLandProjects/lab2$
```

10. Перейдем в графану, включим плагин Zabbix

← → ↻ localhost:3000/plugins/alexanderzobnin-zabbix-app ☆ New Chrome available ⋮

Q Search or jump to... ⌘ ctrl+k + - 🔍 Sign in

Home > Administration > Plugins and data > Plugins > Zabbix

Z

Zabbix

Version

6.0.3

From

Grafana Labs

Dependencies

Grafana >=11.6.0

Signature

Signed Level: Grafana Labs Signed by: Grafana Labs

Uninstall

Disable

Zabbix plugin for Grafana

[GitHub](#) | [Docs](#) | [License](#)

You do not have permission to uninstall this plugin.

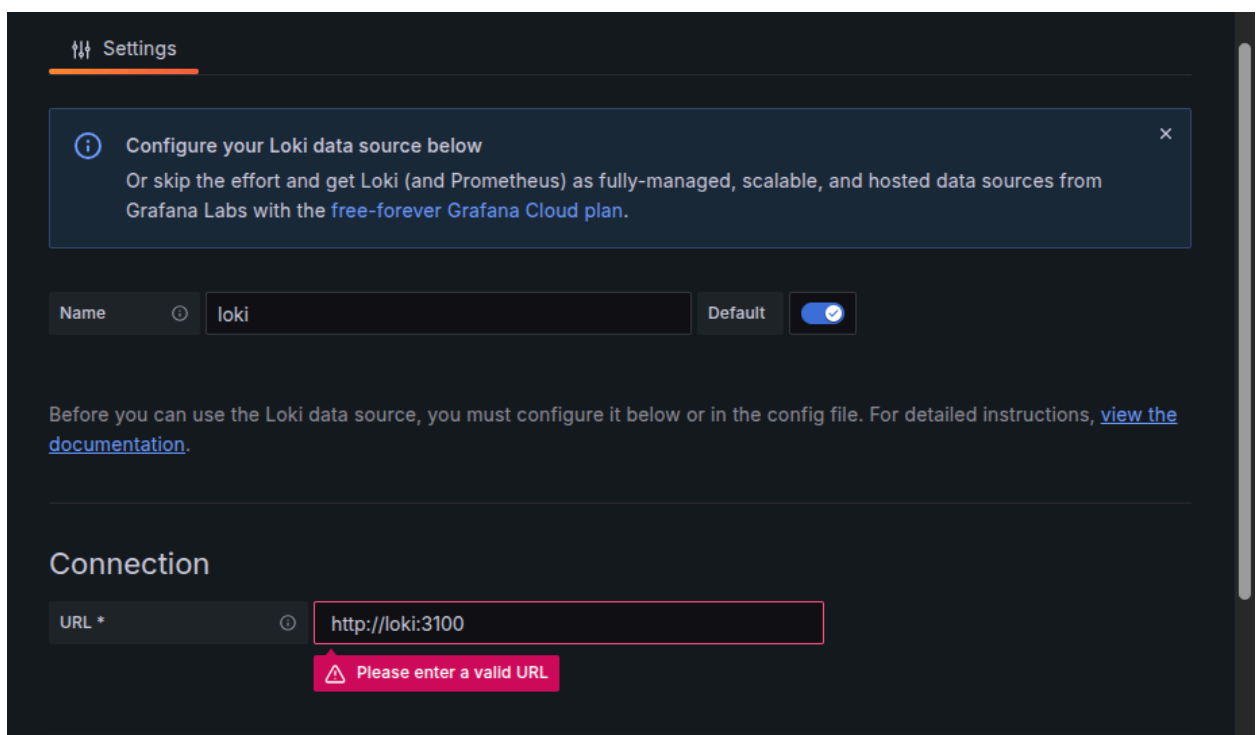
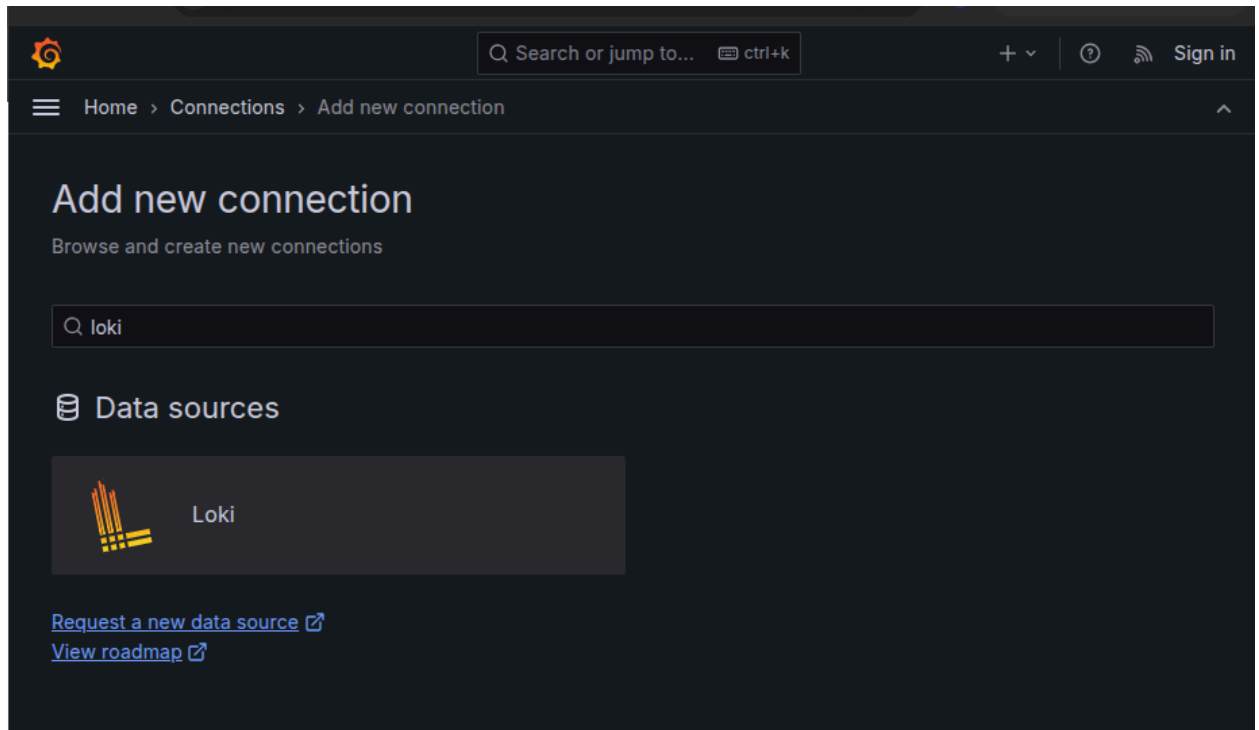
Overview

Version history

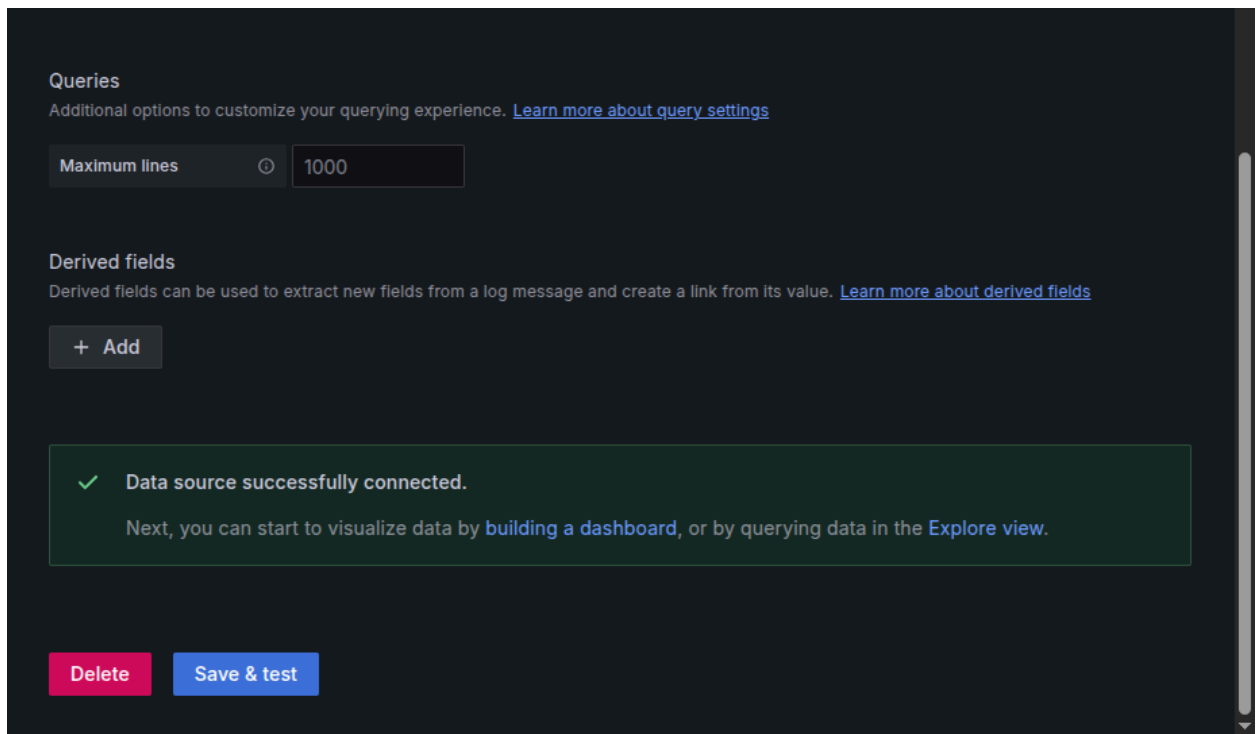
Zabbix plugin for Grafana

version 6.0.3 change log docs latest

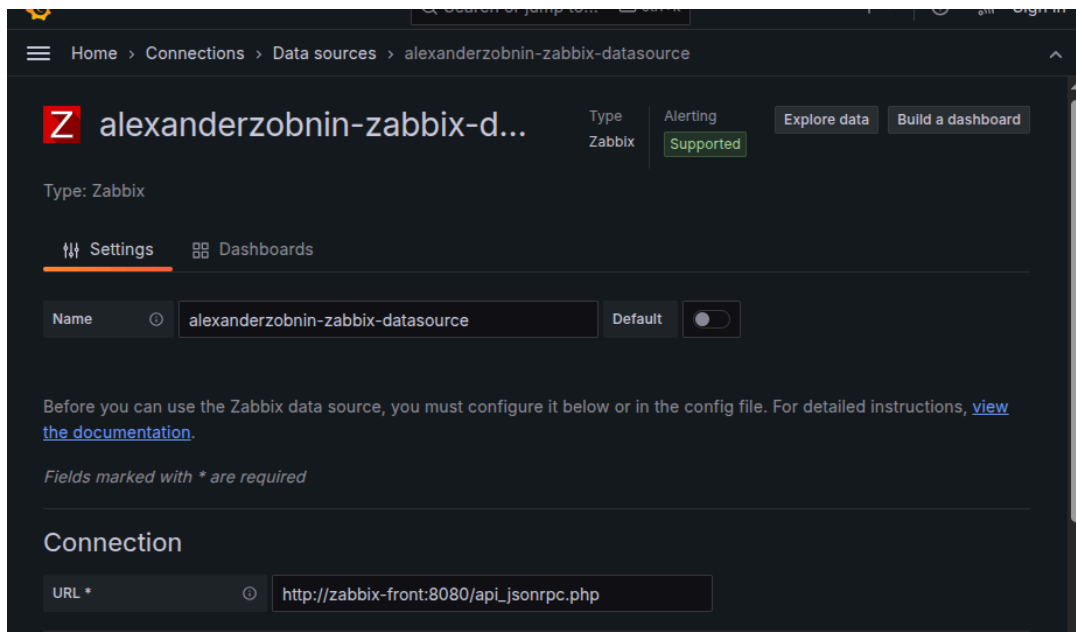
11. Добавим новый Data Source



Загорелось зеленым – датасурс настроен корректно.



12. Аналогично добавим zabbix как data source



Zabbix Connection

Auth type

User and password

Username

Admin

Password

.....

Успех.

Disable data alignment

✓ Zabbix API version: 6.4.21

Next, you can start to visualize data by [building a dashboard](#), or by querying data in the [Explore view](#).

Data sources

+ Add new data source

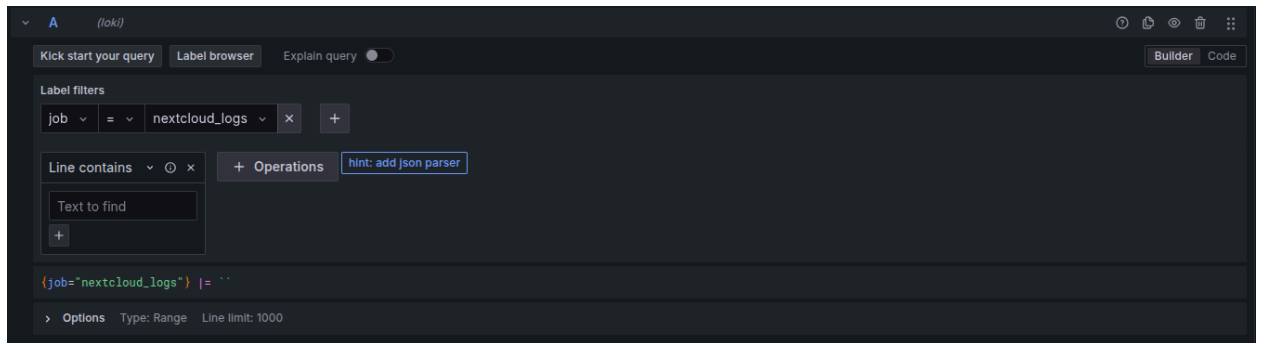
View and manage your connected data source connections

Q Search by name or type

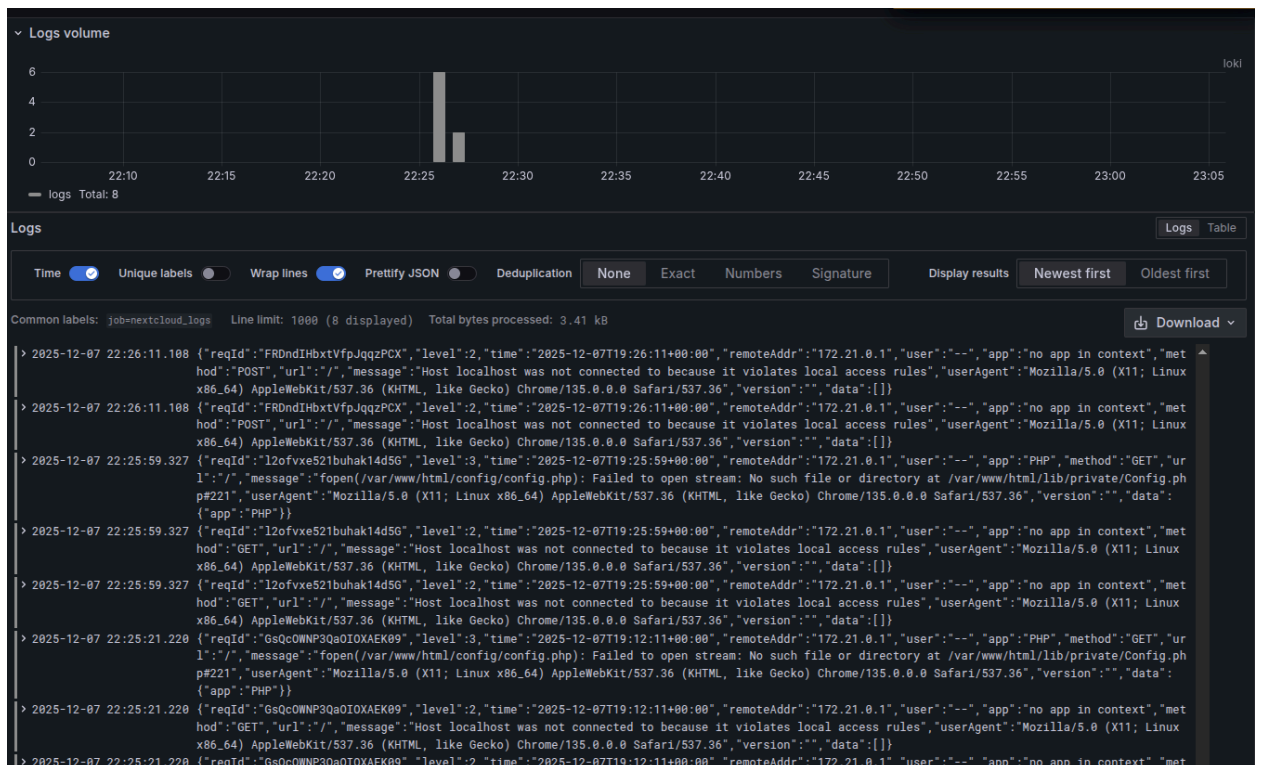
Sort by A-Z

	alexanderzobnin-zabbix-datasource Zabbix http://zabbix-front:8080/api_jsonrpc.php	 Build a dashboard	 Explore
	loki Loki http://loki:3100 default	 Build a dashboard	 Explore

13. Сбилдим PromQL запрос через Builder



Посмотрим на результат его выполнения



14. Построим пару бордов.

1) Логи в виде таблицы в Loki. Фильтр

```
{job="nextcloud_logs"} | = ``
```

тип визуализации – таблица.

The screenshot shows the Grafana interface with a Loki query and a table visualization. The query is `{job="nextcloud_logs"} | = ```. The table visualization displays log entries with columns for labels, time, line, tsNs, and id. The right sidebar shows various visualization options, with the 'Table' option selected.

Panel Title

labels	Time	Line	tsNs	id
{ "filename": "/opt/nc_	2025-12-07 22:26:11.108	{"reqId":"FRDndIHbxtVfpJqqzP	1765135571108971022	1765135571108971022_a2503cb
{ "filename": "/opt/nc_	2025-12-07 22:26:11.108	{"reqId":"FRDndIHbxtVfpJqqzP	1765135571108962644	1765135571108962644_a2503cl
{ "filename": "/opt/nc_	2025-12-07 22:25:59.327	{"reqId":"I2ofvxe52Ibuhak14d5	1765135559327774641	1765135559327774641_2d5b93
{ "filename": "/opt/nc_	2025-12-07 22:25:59.327	{"reqId":"I2ofvxe52Ibuhak14d5	1765135559327760350	1765135559327760350_a07664
{ "filename": "/opt/nc_	2025-12-07 22:25:59.327	{"reqId":"I2ofvxe52Ibuhak14d5	1765135559327754464	1765135559327754464_a07664
{ "filename": "/opt/nc_	2025-12-07 22:25:21.220	{"reqId":"GsQcOWNP3QaOIOX,	1765135521220712251	1765135521220712251_5530454
{ "filename": "/opt/nc_	2025-12-07 22:25:21.220	{"reqId":"GsQcOWNP3QaOIOX,	1765135521220684228	1765135521220684228_5e8d40

Query 1 Transform data 0

Label filters

job = nextcloud_logs

Line contains Text to find

Operations hint: add json parser

{job="nextcloud_logs"} | = ``

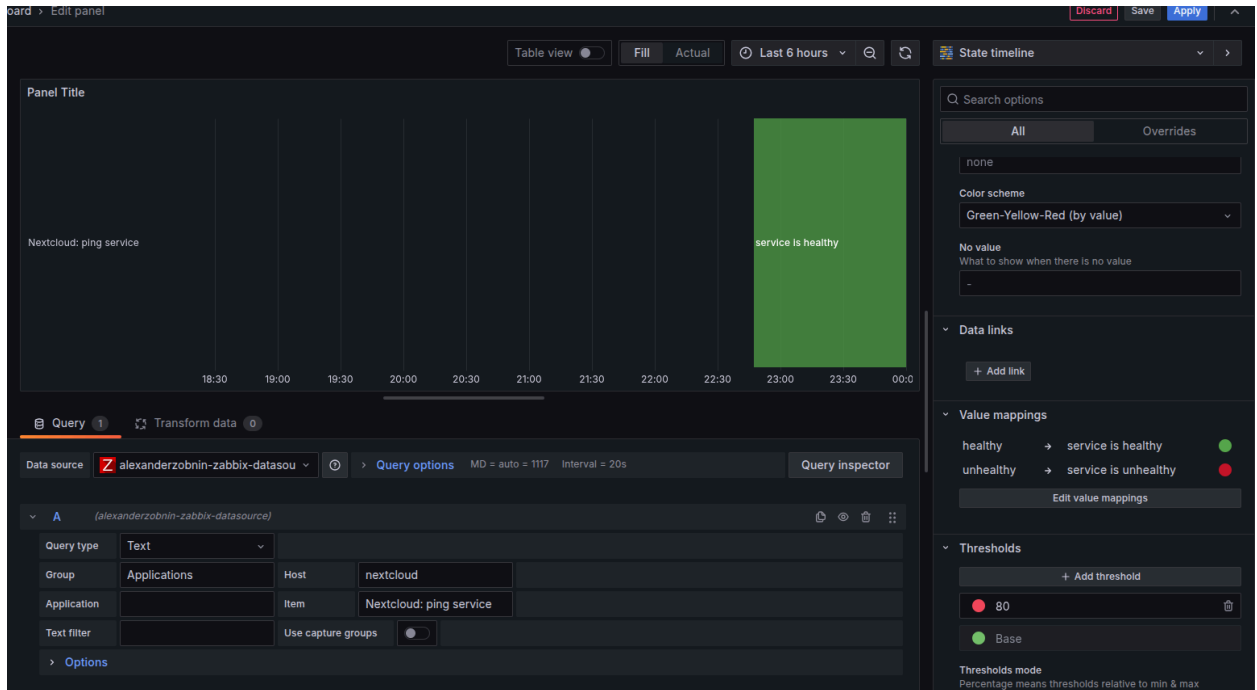
Options Type: Range Line limit: 1000

+ Add query + Expression

Visualizations Suggestions Library panels

- Time series Time based line, area and bar charts
- Bar chart Categorical charts with group support
- Stat Big stat values & sparklines
- Gauge Standard gauge visualization
- Bar gauge Horizontal and vertical gauges
- Table Supports many column styles
- Pie chart The new core pie chart visualization
- State timeline State changes and durations
- Heatmap Like a histogram over time
- Status history Periodic status history
- Histogram Distribution of values presented as a bar chart.
- Text Supports markdown and html content
- Alert list Shows list of alerts and their current status

2) Статус сервера.



15. Ответы на вопросы:

- 1) SLO — это внутренняя цель уровня сервиса, а SLA — официальное соглашение с клиентом с обязательствами.
- 2) Инкрементальный бэкап сохраняет только изменения с последнего любого бэкапа, а дифференциальный — только изменения с последнего полного бэкапа.
- 3) Мониторинг собирает заранее определённые метрики и алерты, а observability позволяет понять поведение системы на основе всех её сигналов без заранее заданных гипотез.